

Tejas Goyal

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EDUCATION

Carnegie Mellon University

Master of Information Systems

Pittsburgh, PA

August 2025 – December 2026

- Courses: LLM Methods and Applications, LLM Systems, AI Agents, Deep Learning

PROFESSIONAL EXPERIENCE

Qualcomm

Machine Learning Engineer Intern

San Diego, CA

May 2026 – Present

- Lifted root-cause success from **54% to 71%** by training a diagnosis agent with GRPO (RL) on execution-grounded rewards over multi-step debugging trajectories in sandboxed AWS EC2 (**3K rollouts**, 8 samples/group)
- Improved node coverage **43%** and cut retrieval noise **37%** over **10K+ logs/day** by fine-tuning a bi-encoder and cross-encoder reranker on mined hard negatives and combining them with BM25 search over a Neo4j knowledge graph
- Served the full daily load from a single on-prem **31B LLM (Gemma 4)** by pre-computing agent branching, shipping the system to production as chat, workflow, and agentic tools with sandboxed, auto-scaling code execution

OpenHands

Machine Learning Engineer

Pittsburgh, PA

August 2025 – Present

- Engineered a configurable timeout system for MCP servers preventing tool execution failures during long-running agent operations, improving reliability for **100,000+ active developers** [PR]
- Built agentic workflow features in the open-source Python agent SDK (tool calling, MCP integration), translating enterprise client requirements into shipped features benchmarked against Cursor, Devin, and Codex

Confluent

Technical Program Manager

Bangalore, India

January 2024 – August 2025

- Drove the roadmap for an AI-powered Customer Intelligence Hub, aligning engineering and go-to-market teams to ship a **40% faster** reporting pipeline for **500+ sales representatives**

Philips

AI Engineer Intern

Bangalore, India

June 2023 – August 2023

- Built a generative AI, LLM-based patient-support application on Azure OpenAI with a RAG pipeline (Pinecone, spaCy chunking), cutting query resolution time **60%** for a projected **\$1.1M** in savings

RESEARCH EXPERIENCE

TEEL Lab | CMU (Advisor: Prof. Jaromir Savelka)

December 2025 – Present

- Trained Qwen2.5-7B tool agents with SFT on filtered teacher trajectories, then GRPO with policy-violation penalties on τ^2 -bench, holding rule families out of every training channel; showed restraint transfers to unseen rules but escalation collapses (**99% → 8%**), a failure violation-rate metrics miss (*ICLR 2027, under review*)
- Proposed **spec learning**, a four-stage compiler that distills **20 preference pairs** into a natural-language system prompt for inference-time alignment with no weight updates; outperforms LoRA-DPO trained on **1,000 pairs** across **6 domains** with a **75% pairwise win rate** (*NeurIPS 2026, under review*) [paper]
- Benchmarked **8 LLMs** on domain-specific span labeling across **5 datasets**; grounding prompts in structured label definitions lifts exact-match F1 from **1.8% to 57.5%** on expert tasks (*ACL 2026 SRW*) [paper]
- Currently researching rubric-based reward design for GRPO in hard-to-verify domains, including rubric generation from preference pairs

PROJECTS

OpenHands – Configurable MCP Server Timeouts | [Open-Source Contribution](#) (90K stars)

- Implemented configurable timeouts for MCP, preventing tool failures in long horizon coding agents **100,000+ devs**

SGLang – WLFU Radix Cache Eviction Policy | [Open-Source Contribution](#) (28K stars)

- Designed WLFU, a log-frequency-decayed LRU eviction policy for SGLang's radix cache, cutting **mean TTFT 22.8%** and **E2E latency 12.8%** vs. LRU on an H100

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, TypeScript

ML: PyTorch, HF Transformers, NumPy, vLLM, SGLang, LoRA fine-tuning, GRPO/RLHF, embeddings, reranking

MLOps & Cloud: model training, evaluation, deployment, AWS, Azure, Docker, Git, Neo4j, PostgreSQL